
Transforming Economic Models into AI Systems: An Illustration with Load Forecasting

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Economic Machine Learning (EML) Lab

What is the EML Lab?

Mission

- ▶ Build economic models that can operate in an **AI/ML environment**
- ▶ Support **policy analysis, economic decision making, and forecasting**

Three Pillars

- ▶ **Frontier research**: combining econometric theory with machine learning
- ▶ **Student training**: producing workforce-ready talent
- ▶ **Industry partnerships**: real-world data, real-world problems

Leadership

- ▶ **Yoosoon Chang**, Executive Director — Professor of Economics, Indiana University
- ▶ **Joon Y. Park**, Research Director — Professor of Economics & Wisniewsky Professor of Human Studies, Indiana University

The Problem:
**Economic Models in an AI/ML
Environment**

The Objectives

Economics has always been **model-based**.

Motivation and Goal

- ▶ How best to support **economic decision making** on **policy choices**, and **managerial/investment decisions**
- ▶ Do this effectively in a modern **AI/ML environment** to support it at the speed and scale real-time decision making demands

Model Requirements for AI Deployment: **ART**

- ▶ **Adaptability** to accommodate heterogeneities across individuals and time-varying institutions
- ▶ **Robustness**: to minimize frequency and magnitude of the risk of system failure
- ▶ **Transparency** to allow for fully automated operation supporting self-diagnosis and self-revision

The Tension

Standard ML Does Not Work for Economic Data Analysis

Challenges

- ▶ Economic data are **very limited**, in contrast with the big data regime standard ML thrives in
- ▶ Economic relationships are **time-varying** due to institutional changes, and **heterogeneous** across agents

Failure of Standard ML Approach

- ▶ Economic relationships are generally **weak** due to **heterogeneous human behaviors** and **time-varying institutions** while need to be trained with only limited amount of data
- ▶ The standard AI and ML approach were developed to infer **strong** relationships assuming affluent training data
- ▶ Off-the-shelf ML tools **grossly fail** if they are applied to analyze economic data

The Itron Lesson: Evidence From the Field

Current Situation

Itron — the most prominent consulting company serving system operators, including MISO and PJM, and the majority of utility companies — studied a wide range of approaches including (i) **regression** (OLS, ridge, lasso, elastic net, quantile, etc.), (ii) **decision tree regressors**, and (iii) **neural network models**

Their Finding

- ▶ Neural networks have the best **in-sample** fit
- ▶ But they generally **fail out-of-sample**
- ▶ So they fall back to **simple OLS regression**

EML's Position

- ▶ Simple OLS regression is **not** the answer either
- ▶ The problem demands **customized ML grounded in economic theory**

EML's Approach: Three Stages of Model Building

We build economic models for AI/ML environments in **three progressive stages**:

Stage 1 Conventional economic model, **simplified** to be estimable by existing econometric methods. *No ML used.*

Stage 2 The same economic model, now estimated using **customized ML** — no simplification needed. *More flexible than Stage 1.*

Stage 3 The Stage 2 model extended to incorporate **new and different information** that Stage 2 could not accommodate. *More general than Stage 2.*

Most economics researchers still operate at **Stage 1**. Getting to Stage 2 requires careful, non-trivial customization of ML tools — which is why few have done it successfully. **EML has passed Stage 2** and is now actively working in Stage 3.

Three Stages of Model Building

Stage 1: The Conventional Economic Model

What It Is

- ▶ A theory-grounded model — structural or reduced form
- ▶ Built from **economic theory**, prior knowledge, and accumulated empirical regularities
- ▶ **Not** built from data alone

The Challenge

- ▶ To estimate with existing econometric methods, the model must be **simplified**: functional forms restricted, relationships linearized, dimensionality reduced
- ▶ These simplifications **sacrifice flexibility** and may miss important **nonlinearities** and **interactive effects**

Where Most Research Stops

- ▶ Stage 1 models are rigorous, interpretable, and grounded
- ▶ But they leave real **explanatory power** on the table

Stage 2: Training the Model with Customized ML

The Key Advance

- ▶ With customized ML, we **no longer need to simplify** the economic model to make it estimable, even with limited data
- ▶ ML provides the **flexibility** to estimate the original, unsimplified model directly

The Toolkit

- ▶ Random forests, SVM, RKHS methods, functional data analysis
- ▶ Each tool **customized** to respect the theoretical structure — not applied off-the-shelf
- ▶ The knowledge of economic theory and accumulated empirical experience is **distilled** into this Stage 2 model

Stage 2: Scope and Limitations

What Stage 2 Can Accommodate

- ▶ Accommodates only identified covariates for which **usable data exist** — e.g., total load, temperature, calendar variables

What Stage 2 Cannot Do

- ▶ Cannot incorporate new information sources that don't fit naturally into the conventional model structure — e.g., **textual signals** extracted via large language models (LLMs)
- ▶ These restrictions, however, are what make Stage 2 **estimable, interpretable, and reliable**

Stage 3: Obtaining a Flexible Transformer

The Motivation

- ▶ Stage 2 is strong and interpretable — but bounded by what could be specified in a conventional framework
- ▶ New information keeps arriving: **unstructured data**, real-time signals, textual sources (e.g., market uncertainty captured via LLM-based analysis)
- ▶ Challenge: how to **add new information** to the Stage 2 model, and how best to **use the combined information**?

The Approach

- ▶ Use the Stage 2 model as a **data-generating process** to produce rich synthetic data
- ▶ Train a flexible **transformer** on this data, using constraints motivated by economic theory and empirical experience
- ▶ The resulting transformer inherits the knowledge **distilled in Stage 2** and is more general — capable of incorporating new information Stage 2 could not accommodate

External Interpretability

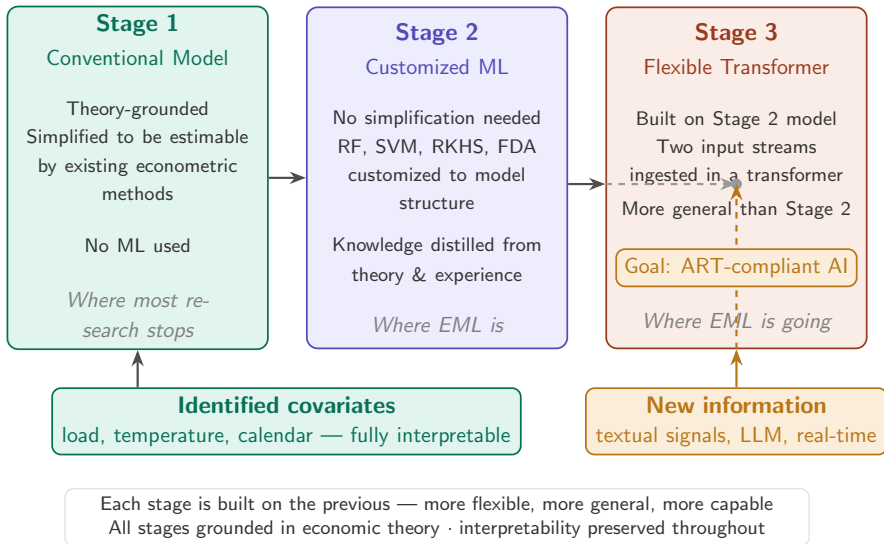
What Is and Is Not Interpretable

- ▶ The Stage 3 transformer's internal mechanics are not directly interpretable — true of all deep learning
- ▶ What **is** interpretable: inputs (defined by the economic model), outputs (policy-relevant quantities), and rules and behavioral patterns (benchmarked against Stage 2)
- ▶ Policymakers can understand **what goes in, what comes out, and why** — without seeing inside

How This Differs from a Black Box

- ▶ The Stage 3 transformer functions as a computationally efficient **approximator** of an economically disciplined model
- ▶ The black box is real — but it is **limited and controlled** on all sides by economic structure

EML's Approach: Three Stages of Model Building



Relationship to Existing Work

How EML Relates to and Differs from Existing Approaches

- ▶ **PINNs**: analogous logic, but physical laws vs. probabilistic economic theory — richer and harder
- ▶ **Frontier ML in economics literature**: shares Stage 2 commitment, much less effectively than us, and stops there — no distillation, no Stage 3
- ▶ **Standard knowledge distillation**: NN $\rightarrow$ smaller NN; EML distills economic model using constraints motivated by economic theory and empirical experience — different purpose and structure
- ▶ **Agent-based models**: share synthetic data interest, but lack Stages 2–3

To the best of our knowledge, the full three-stage pipeline as an integrated system does not have a direct precedent.

Illustration: EML Load Forecaster

Why Load Forecasting?

30-Year Track Record

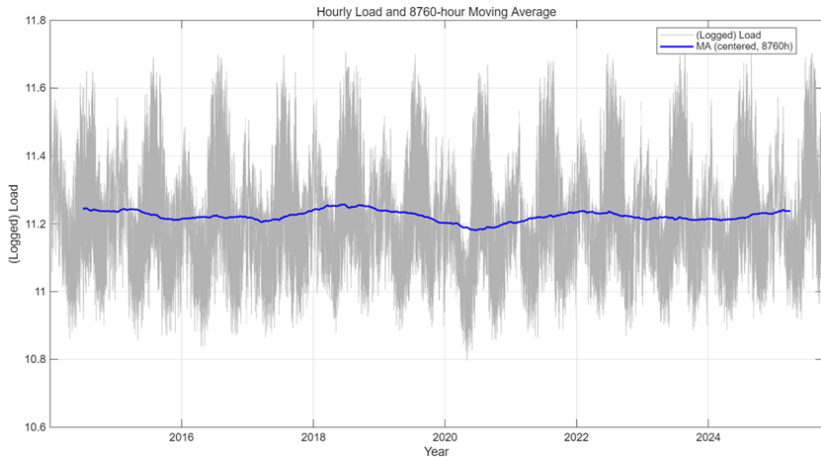
- ▶ 1995–2019: Numerous projects with Korea Power Exchange (KPX), Korea Electric Power Corporation (KEPCO), and Korea Gas Corporation (KOGAS) — over \$5M in total
- ▶ 2019–present: Co-founders of iMAES, providing energy forecasting solutions in Korea
- ▶ Multiple publications in *Energy Economics*

Why It Matters for Decision Making

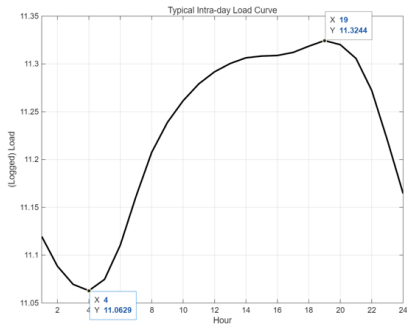
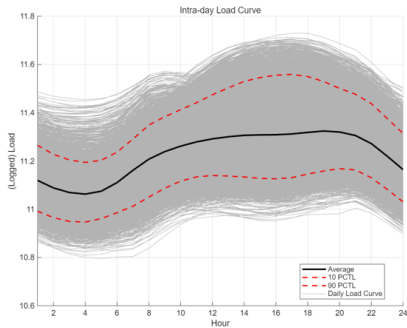
- ▶ **System operators**: day-ahead forecasts for dispatch and grid reliability
- ▶ **Utility companies**: procurement, hedging, IRP
- ▶ **Traders**: position-taking in energy markets
- ▶ Stakes are rising fast — electrification, data centers, renewable integration

The EML load forecaster illustrates the three-stage pipeline concretely, with **Stages 1 and 2 fully operational**.

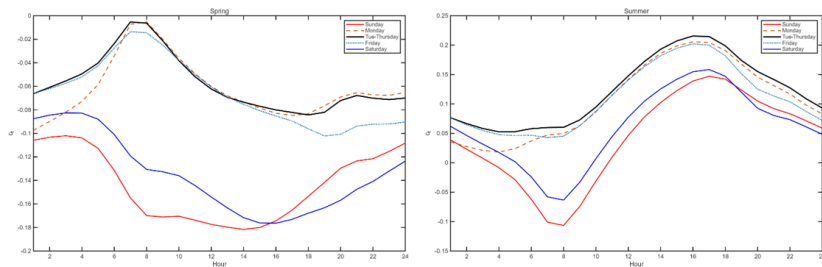
Stylized Facts of Load: Time-Varying Trend



Stylized Fact: Intra-Day & Cross-Day Variation

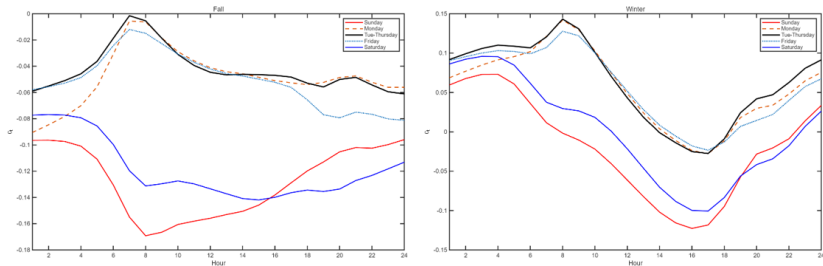


Stylized Fact: Season-Dependent Daily Patterns



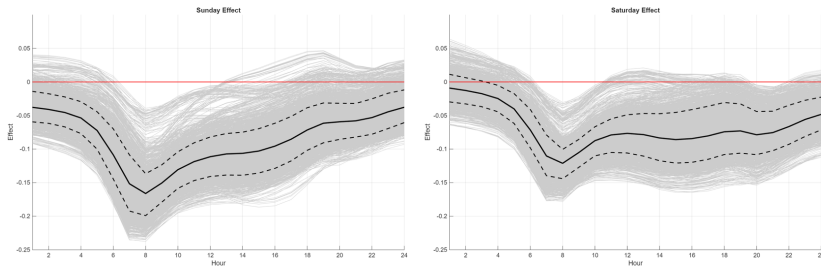
- ▶ In spring, weekday peaks occur in the morning, and the gap between weekdays and weekends is pronounced.
- ▶ In summer, load peaks around 18:00 regardless of day-of-week.

Stylized Fact: Season-Dependent Daily Patterns



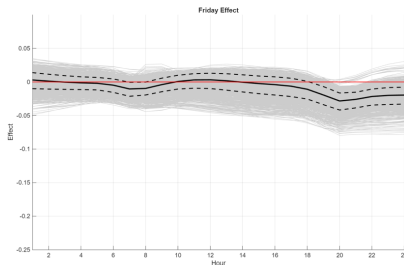
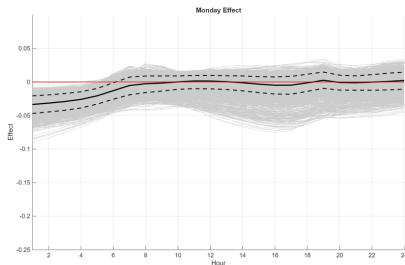
- ▶ Fall mirrors spring: morning weekday peaks and a large weekday–weekend gap.
- ▶ Winter: similar overall shapes, but different peak timing.
 - ▶ Weekday peak around 8 AM;
 - ▶ Weekend peak around midnight.

Estimated Day-of-Week Effects



- ▶ Effects are estimated relative to mid-week (Tue–Thu).
- ▶ Sunday and Saturday exhibit consistently lower load across most hours.

Estimated Day-of-Week Effects



- ▶ Monday shows lower load in the morning (Sunday spillover effect).
- ▶ Friday shows lower load in the afternoon (Saturday anticipation effect).

Stage 1 in Action: The Economic Model of Load

Functional Data Approach

- ▶ The load curve is treated as a **functional object**
- ▶ Each day's 24-hour load profile is a curve, not a vector of scalars

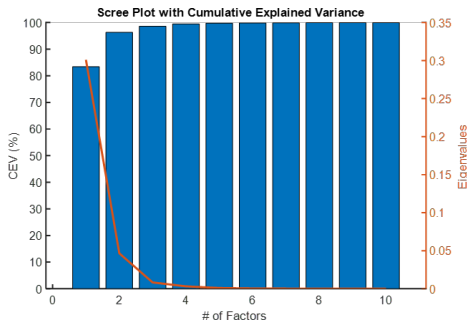
Model Decomposition

$$\ell_t = \tau_t + c_t + f_t$$

- ▶ (τ_t) : **trend component** — long-run structural shifts in demand
- ▶ (c_t) : **cyclical component** — temperature and calendar-driven variation
- ▶ (f_t) : **functional autoregressive (FAR) component** — intra-day load curve dynamics

Each component has clear **economic interpretation** and is estimated separately. The decomposition structure is not a statistical choice — it reflects how economists understand the drivers of electricity demand.

FPCA on Daily Load Curves

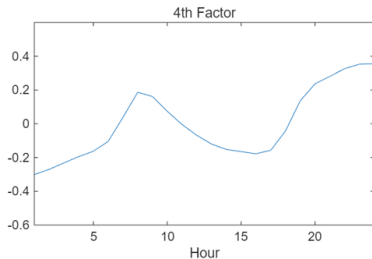
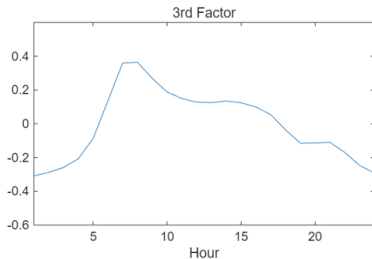
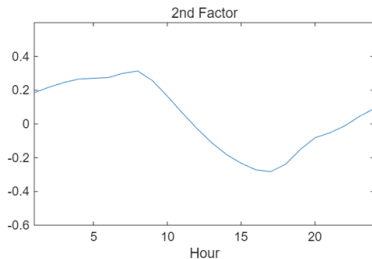
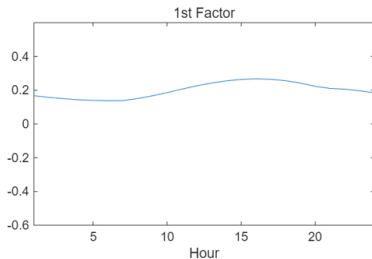


Factor	CEV
1	83.42
2	96.33
3	98.61
4	99.49
5	99.76
6	99.86
7	99.91
8	99.94

Eight factors explain virtually all variation — 99.94%.

First Four Functional Principal Components

Functional Principal Components



The Trend Component (τ_t)

What It Captures

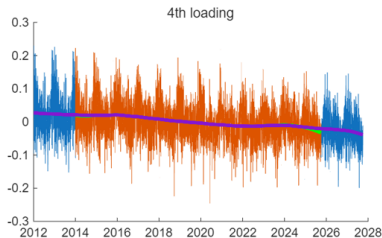
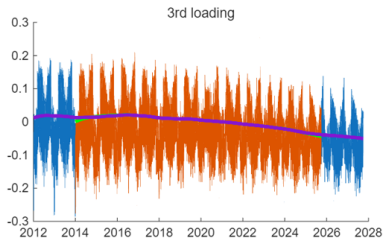
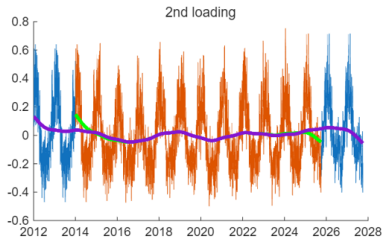
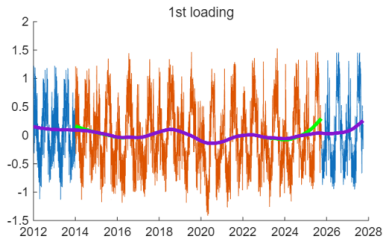
- ▶ Long-run structural shifts in electricity demand
- ▶ Efficiency improvements, electrification, industrial composition changes

Estimation

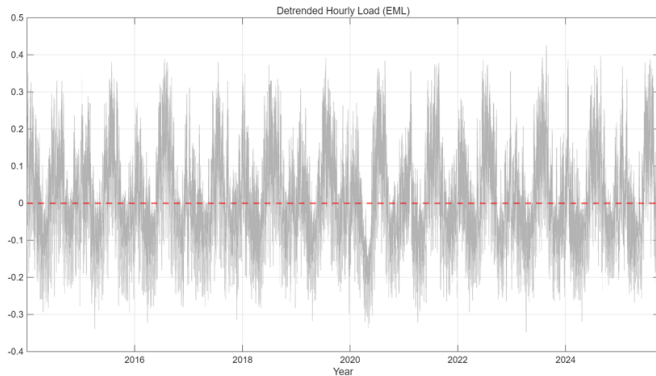
- ▶ Load curve represented via **Functional Principal Component Analysis (FPCA)**
- ▶ Trend extracted from each FPCA loading via **smoothing splines**
- ▶ A Stage 1 operation: a well-established nonparametric method within the conventional econometric framework

Trend Estimation for the First Four Loadings

Preliminary Trend ($\lambda = 0.95$)

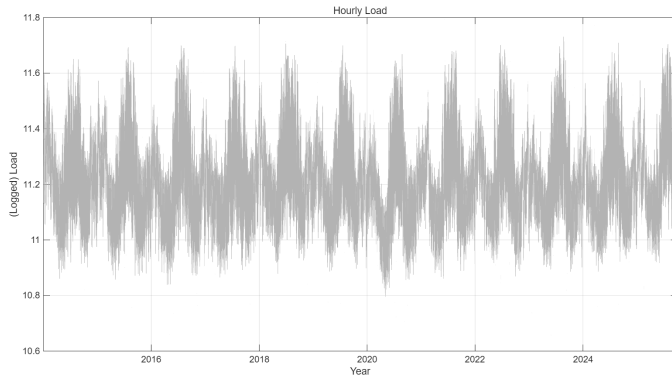


Detrended Hourly Load



- ▶ After trend removal, the series exhibits much smaller variation.
- ▶ This improves identification of cyclical dynamics (temperature, calendar, and persistence).
- ▶ Current setting: $\lambda = 0.95$; we can consider smaller values as intra-year variation is further reduced.

Recall: Hourly Load



The Cyclical Component (c_t): Model Structure

Functional Coefficient Model

$$c_{it} = \alpha_i(t, \mu_t, \nu_t, \delta_t) + \beta_i(t, \mu_t, \nu_t, \delta_t) c_{i,t-1} + \varepsilon_{it}$$

- ▶ Coefficients vary as functions of:
 - ▶ t : time (slow structural evolution)
 - ▶ μ_t : short-term temperature (spot effect)
 - ▶ ν_t : cumulative temperature (longer-term effect)
 - ▶ δ_t : calendar effects
 - ▶ c_{t-1} : persistence of cyclical dynamics
- ▶ Structure dictated by **economic understanding** of how electricity demand responds to weather and calendar patterns — not chosen for statistical convenience

In Stage 1, estimating this requires **simplifying assumptions** about functional forms and interactions — leaving flexibility on the table.

The Cyclical Component: Customized ML (Stage 2)

Customized Random Forest

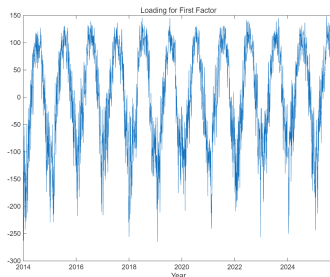
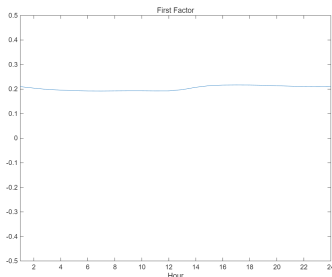
- ▶ We estimate c_t using a **customized random forest** — no simplification needed
- ▶ Two key customizations:
 - ▶ Modified to fit **functional coefficient models** rather than predict scalar outcomes
 - ▶ Modified to allow **local kernel smoothing** over time t for temporal regularity
- ▶ These customizations **improve forecasting performance** substantially — the stark contrast with the Itron finding

Calendar Effects (δ_t): Carefully Engineered

- ▶ Day-of-week: Monday, Friday, Saturday, Sunday
- ▶ Regular holidays: MLK Day, Washington's Birthday, Memorial Day, Juneteenth, Labor Day
- ▶ Special holidays: Thanksgiving, Christmas, New Year's Day
- ▶ Special event days: **Super Bowl Sunday**, **General Election Day**

Short-Term Temperature (μ_t)

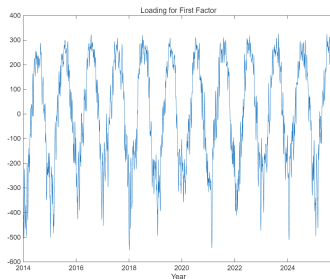
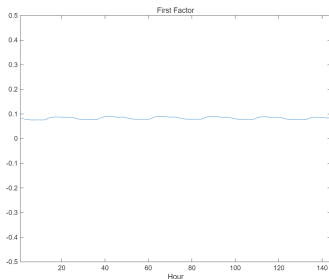
- ▶ Construct daily 24-hour temperature paths.
- ▶ Hourly temperature data from 11 states covered by MISO, aggregated using population weights.
- ▶ Apply FPCA to daily temperature curves.
- ▶ μ_t : loading corresponding to the first FPC.
- ▶ The first FPC captures the daily average temperature effect.



Temperature is treated as a functional object.

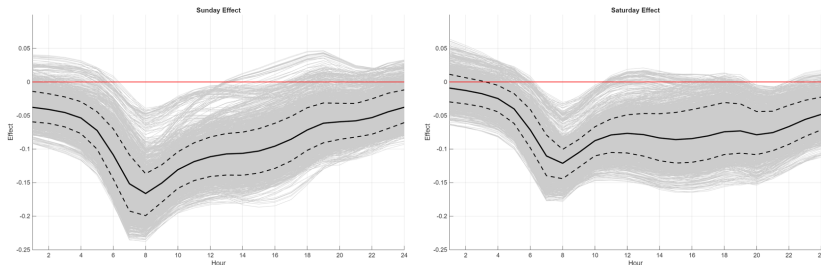
Cumulative Temperature (ν_t)

- ▶ ν_t : cumulative temperature for long-run effects (thermal inertia / persistence).
- ▶ Construct a 144-hour (past 6 days) temperature path using MISO 11-state population-weighted hourly data.
- ▶ Apply FPCA to these 144-hour temperature curves.
- ▶ ν_t : loading corresponding to the first FPC.



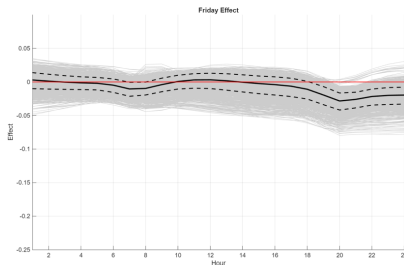
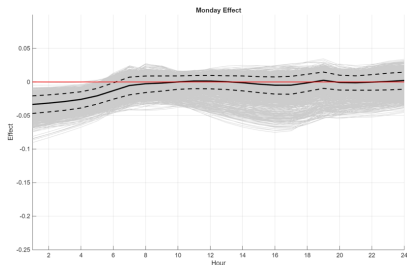
Captures accumulated heating/cooling conditions.

Recall: Estimated Day-of-Week Effects



- ▶ Effects are estimated relative to mid-week (Tue–Thu).
- ▶ Sunday and Saturday exhibit consistently lower load across most hours.

Recall: Estimated Day-of-Week Effects



- ▶ Monday shows lower load in the morning (Sunday spillover effect).
- ▶ Friday shows lower load in the afternoon (Saturday anticipation effect).

Calendar Effect Design: Day-Type Classification (δ_t)

- ▶ Load depends not only on the current day, but also on whether the previous and next days are holidays.
- ▶ We classify each day into 8 types based on (Previous Day, Today, Next Day).

Type	Previous Day	Today	Next Day
Type Mon (1)	Holiday	Weekday	Weekday
Type Work (2)	Weekday	Weekday	Weekday
Type Fri (3)	Weekday	Weekday	Holiday
Type Sat (4)	Weekday	Holiday	Holiday
Type Sun (5)	Holiday	Holiday	Weekday
Type MWH (6)	Weekday	Holiday	Weekday
Type Sand (7)	Holiday	Weekday	Holiday
Type MD (8)	Holiday	Holiday	Holiday

Captures spillover and anticipation effects.

The FAR Component and Full Forecast Assembly

Functional Autoregressive (FAR) Component

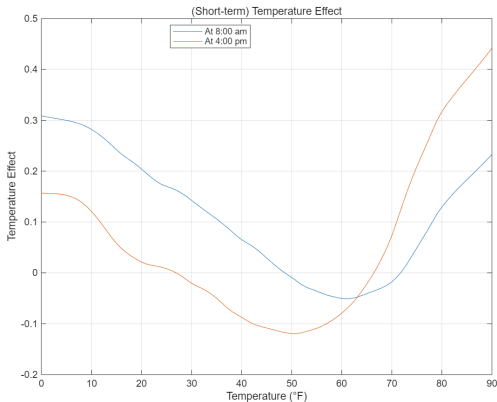
- ▶ Captures **intra-day load curve dynamics** not explained by trend or cyclical components
- ▶ Specified as a first-order functional autoregression:
$$f_t = Af_{t-1} + \varepsilon_t$$
- ▶ Estimated as a **VAR on functional principal components** (8 loadings)
- ▶ Adds approximately **0.05% improvement** in MAPE — modest but real

Full Forecast Assembly

- ▶ Three components estimated separately and combined:
$$\hat{\ell}_t = \hat{\tau}_t + \hat{c}_t + \hat{f}_t$$
- ▶ **Modularity** is a feature: each piece is interpretable and diagnosable independently
- ▶ Fully **automated**: runs without human input for the entire forecast period

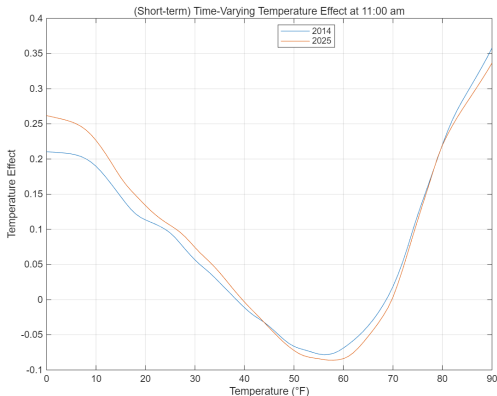
Diagnosis

Nonlinear Temperature Effects by Hour



- ▶ Temperature effects are highly nonlinear.
- ▶ Strong hour-of-day heterogeneity:
 - ▶ At 8 AM: strong heating effect (cold-driven demand).
 - ▶ At 4 PM: dominant cooling effect (heat-driven demand).
- ▶ Afternoon demand shows larger overall sensitivity.

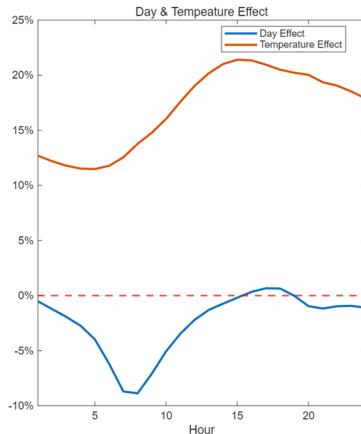
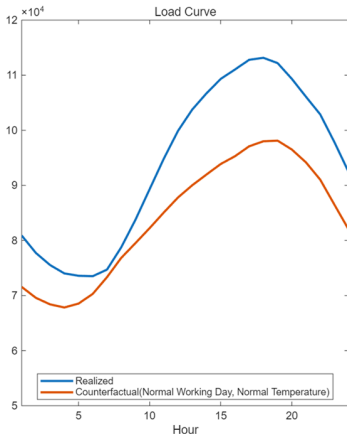
Time-Varying Temperature Sensitivity (11 AM)



- ▶ 2014 vs 2025 (11 AM): stronger cold sensitivity, weaker heat sensitivity.
- ▶ Possible drivers: heating electrification; improved cooling efficiency; DR/peak management.

Counterfactual Decomposition: Heat Shock

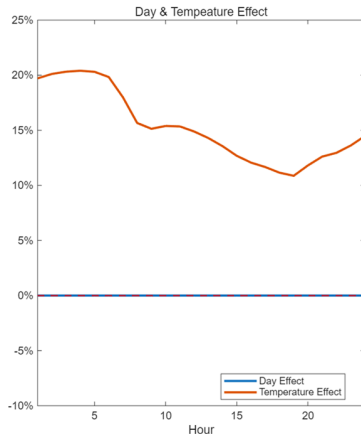
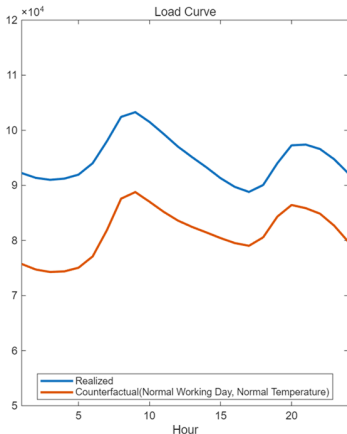
Case Study – June 21, 2025 (Saturday)



- ▶ 24h avg temp: 83°F (NY normal: 75°F) $\Rightarrow +9^{\circ}\text{F}$ shock
- ▶ 144h avg temp: 74°F (NY normal: 74°F) $\Rightarrow$ no cumulative anomaly
- ▶ Counterfactual baseline: *normal working day + normal temperature*

Counterfactual Decomposition: Persistent Cold Shock

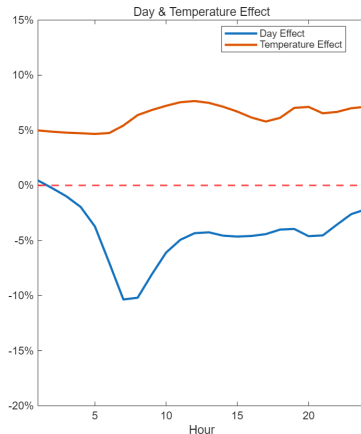
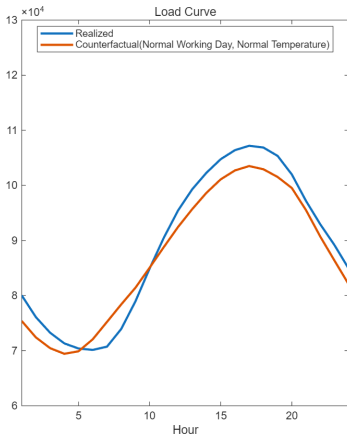
Case Study – February 20, 2025 (Thursday)



- ▶ 24h avg temp: 14°F (NY normal: 35°F) $\Rightarrow -21^{\circ}\text{F}$ shock
- ▶ 144h avg temp: 21°F (NY normal: 33°F) $\Rightarrow -13^{\circ}\text{F}$ cumulative anomaly
- ▶ Day effect ≈ 0 (regular weekday)

Counterfactual Decomposition: Independence Day

Case Study – July 4, 2025 (Friday)



- ▶ 24h avg temp: 81°F (NY normal: 76°F) $\Rightarrow$ +5°F shock
- ▶ 144h avg temp: 78°F (NY normal: 75°F) $\Rightarrow$ +3°F cumulative anomaly
- ▶ Large negative holiday effect offset by positive temperature effect

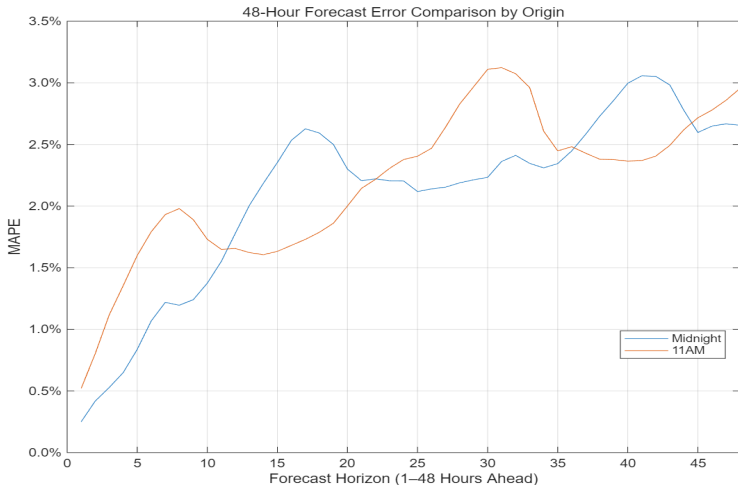
Forecasting Performance

Evaluation Period: Jan 1 – Sep 30, 2025 (272 days)

	One-day Ahead (MAPE)		Two-day Ahead (MAPE)	
	Midnight	11AM	Midnight	11AM
Normal Days (245)	1.69%	1.73%	2.67%	2.67%
Special Days (27)	1.50%	1.54%	2.54%	2.54%
Overall (272)	1.67%	1.71%	2.54%	2.66%

- ▶ One-day-ahead MAPE: **1.67%** (Midnight) and **1.71%** (11AM)
- ▶ Special days forecast **better** than normal days (1.50%/1.54% vs. 1.69%/1.73%) — a sign of the careful calendar modeling
- ▶ Fully **automated** system — no human input throughout the test period

MAPE by Forecast Horizon (48 Hours Ahead)



- ▶ Near-term errors are reduced by the auxiliary model.
- ▶ Errors rise gradually with horizon but remain well-contained. Forecast difficulty concentrates around late-afternoon peak hours.

Comparison with MISO and the Road to Stage 3

Comparison with MISO

- ▶ EML's Stage 2 system — automated, using only **publicly available** total load and temperature data — is competitive with MISO's human-supervised operational forecasts
- ▶ MISO has access to disaggregated regional and sectoral data that EML does not yet use

What Stage 3 Will Add

- ▶ Incorporate **new information sources**: sectoral loads, textual signals, real-time data streams
- ▶ Preserve the knowledge **distilled in Stage 2** as the core of the system
- ▶ The three-stage progression is not just a research program — it is an **engineering roadmap** for a fully **ART**-compliant AI forecasting system

EML Load Forecaster: Live Dashboard

Live Forecasting Dashboard

- ▶ [\[Live Dashboard Link\]](#)

The dashboard displays day-ahead forecasts together with **counterfactual decompositions** of day-type and temperature effects, reinforcing our core principle:

*A forecasting system should not only **predict**, but also **explain**, **diagnose**, and **improve itself** in real time.*

EML Lab and Closing

EML Lab: Who We Are

Lab Directors

- ▶ Yoosoon Chang, Executive Director — Professor of Economics, Indiana University
- ▶ Joon Y. Park, Research Director — Professor of Economics, Indiana University

Student Fellows

- ▶ Kwanpyo Ko, Ralph Fernando, Wooseok Sim, Sangmyung Ha, Ali Zarifhonarvar

Industry Engagement

- ▶ Pilot forecasting programs: MISO, CAISO, PJM, Indiana utilities (AES, NIPSCO)
- ▶ 2025 EML Quant Conference, 2026 EML Energy Conference

Broader Research

- ▶ Macroeconomic forecasting (St. Louis Fed), Inequality & mobility (U Chicago), Finance (KAIST), Health economics (IU O'Neill School)

Summary

Stage 1 Conventional economic model, simplified to be estimable. Rigorous and interpretable — but restricted. *Where most research stops.*

Stage 2 Customized ML estimates the model without simplification. Knowledge of economic theory and empirical experience is **distilled** into a more flexible, better-performing model. *Where EML is.*

Stage 3 A general and flexible transformer, built on the Stage 2 model, capable of incorporating new and different information. *Where EML is going.*

We are not applying AI to economics.

We are transforming economic models into AI systems.

Thank You

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